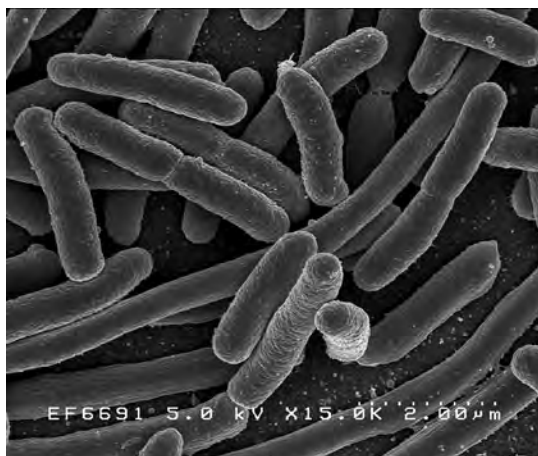


- 2 The photograph shows bacterial cells of *E.coli*, as seen using an electron microscope.



(Source: © BSIP SA/Alamy Stock Photo)

The growth of bacteria in a liquid culture can be investigated.

The number of bacterial cells in the culture can be measured using the dilution plating technique.

- (a) Identify **one** risk that might be encountered when carrying out this technique and how this risk could be reduced.

(2)

Risk

.....

.....

How this risk could be reduced

.....

.....

.....

- (b) The growth rate constant (k) for a bacterial population can be found using the equation:

$$k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$$

N_t is final number of bacteria

N_0 is the initial number of bacteria

t is the growth period in hours

The initial number of bacterial cells was 900.

After 2 hours, there were 14 000 bacterial cells in the culture.

Calculate the growth rate constant, k .

Give your answer to **three** significant figures.

(3)

Answer

(c) (i) State **two** abiotic variables that could affect the growth rate of these bacteria.

(2)

First variable

Second variable

(ii) Choose **one** of the variables you have identified in (c)(i).

Describe how this variable could be controlled and the effect it could have on the results if it is not controlled.

(2)

Variable

Describe how this variable is controlled.

Describe the effect this variable could have on the results if it is not controlled.

(Total for Question 2 = 9 marks)